

Nvidia Final Report (Part 2)

Token Lifecycle and Entropy Analysis in LLMs

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Executive Summary

In this study, we focus on analyzing token entropy across four distinct LLM architectures: Transformers, Mamba State Space Models, Diffusion, and Mixture-of-Experts. This document serves as a summary to the insights we get on the entropy of different LLM architectures. A general algorithm will also be provided so that readers can easily analyze any open weight LLMs of their choice.

Models are trained and tested on the Wikitext dataset unless otherwise stated.

What is Token Entropy?

“Entropy” can be loosely understood as randomness. In this context, “token entropy” refers to how hard tokens are to guess. The higher the entropy, the harder it is to guess the token. Token entropy is calculated using Shannon Entropy, which has the formula:

$$H(x) = - \sum_{i=1}^n P(x_i) \log_b P(x_i)$$
 where $H(x)$ is the Shannon Entropy and $P(x_i)$ is the probability of specific outcome x_i happening.

Models of the same architecture do not have the same entropy, but models of the same family have similar entropy profiles

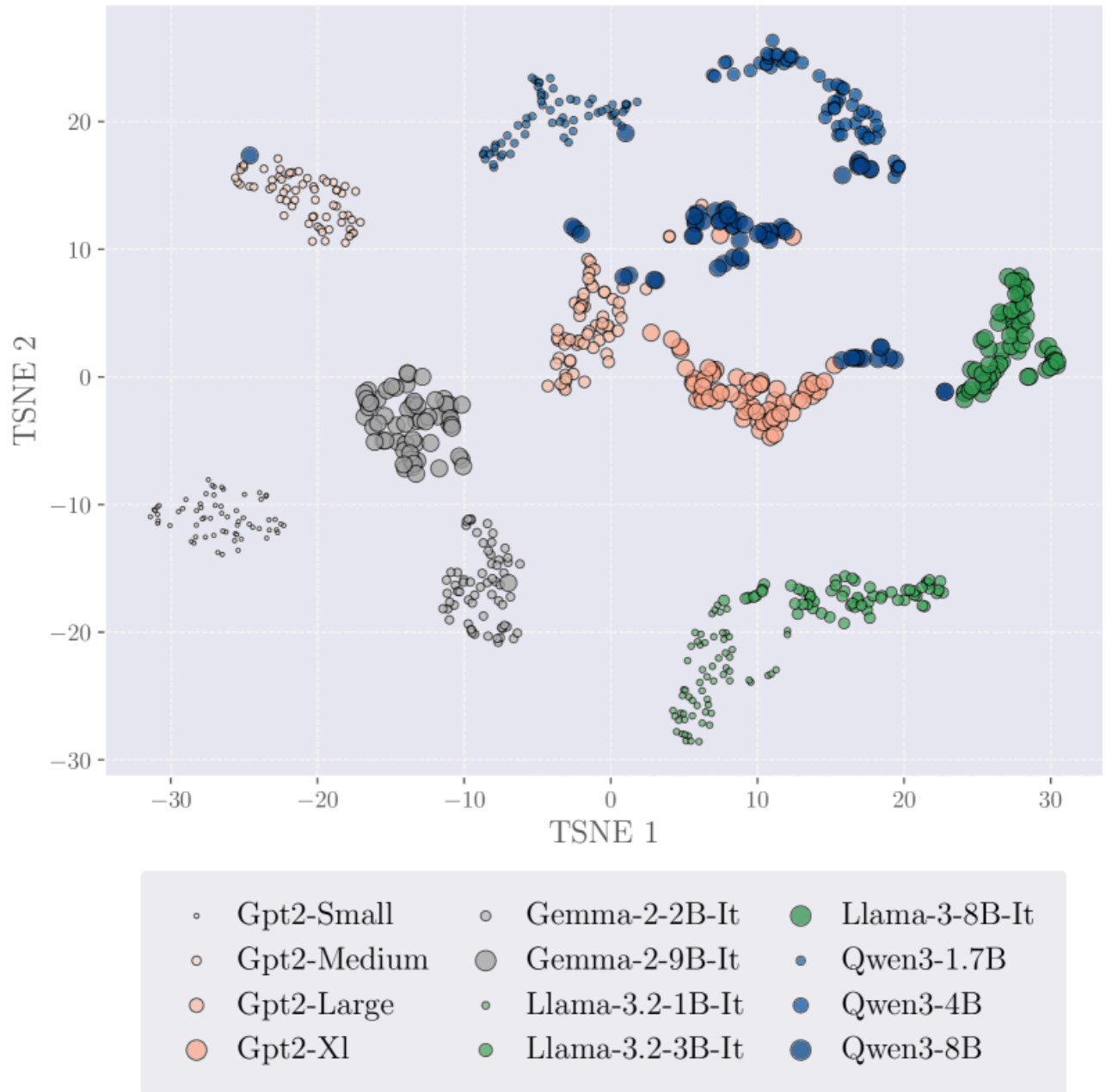


Figure 1: t-SNE projection of aggregated entropy profiles obtained from a blank prompt for different model families. Profiles cluster by family rather than by model size, indicating family-specific token prediction dynamics. Point size scales with number of parameters.

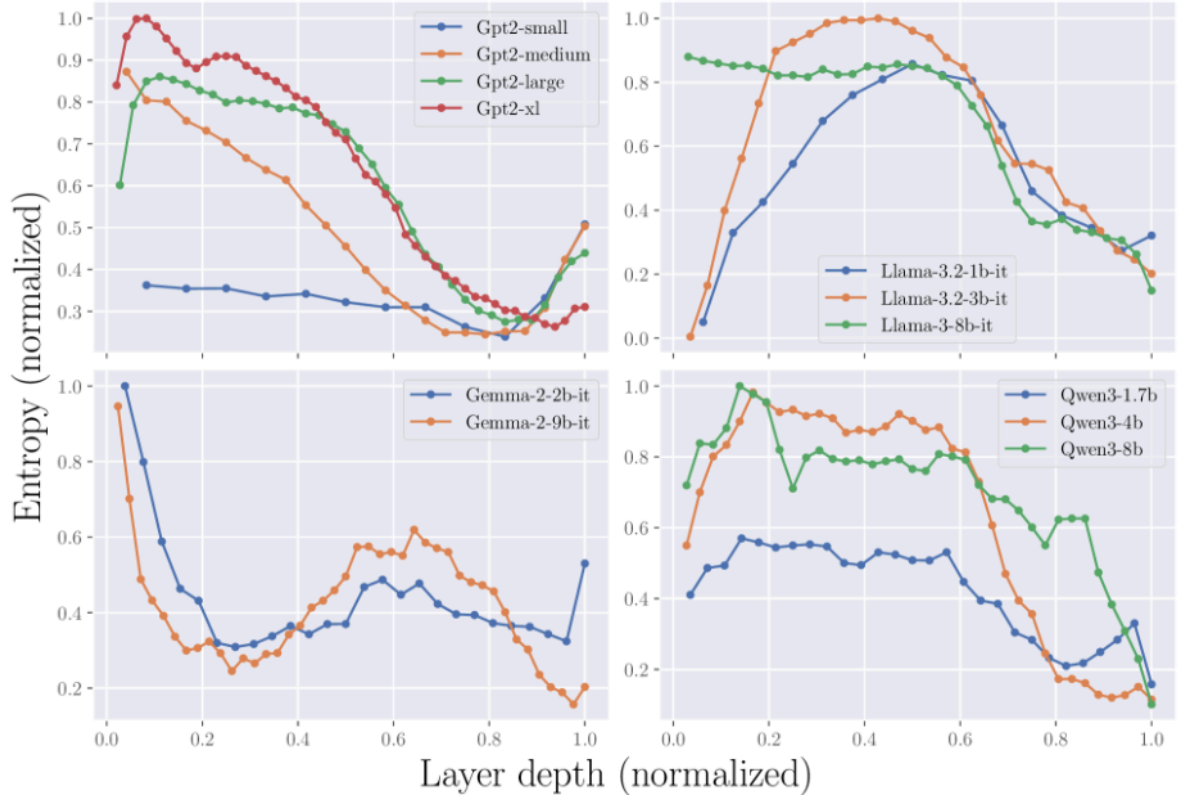


Figure 2: Average entropy profiles over 32 generated tokens for different model families. When depth is normalized, models within the same family exhibit aligned entropy trajectories, revealing family-specific coarse-to-fine token prediction dynamics.

Ali [et.al.](#) (2026) found that:

Entropy profiles cluster by model family rather than by size (Fig. 1). In a low-dimensional projection, models from the same family form coherent groups, while distinct families remain well-separated. This indicates that prediction dynamics are not idiosyncratic to individual checkpoints, but reflect a shared computational structure induced by architectural and training choices.

Qualitatively, different families exhibit distinct entropy regimes (Fig. 2). GPT models start with high entropy and gradually sharpen their predictions, while Llamas display low-entropy initial stages followed by extended high-entropy plateaus and a final refinement. These differences suggest that model families combine expansion and pruning strategies in systematically different ways.

Depth-rescaling invariance and coarse-to-fine behavior. A remarkable regularity emerges when comparing models of different scales in the same family. While absolute depth varies substantially, entropy profiles exhibit a striking alignment when mapped to a relative layer index (Fig. 2). This suggests that model scale primarily dictates the resolution of the computation rather than its qualitative form. From this perspective, smaller models (less than 4B parameters) appear to implement a coarser discretization of the token prediction

dynamics found in their larger counterparts: additional layers serve to refine the signal without fundamentally altering the underlying sequence of expansion and pruning

Evidently, token entropy is similar for models of the same family. It can be also inferred that despite being of the transformer architecture, these models do not share similar entropy profiles. Further evidence can also be found as shown below:

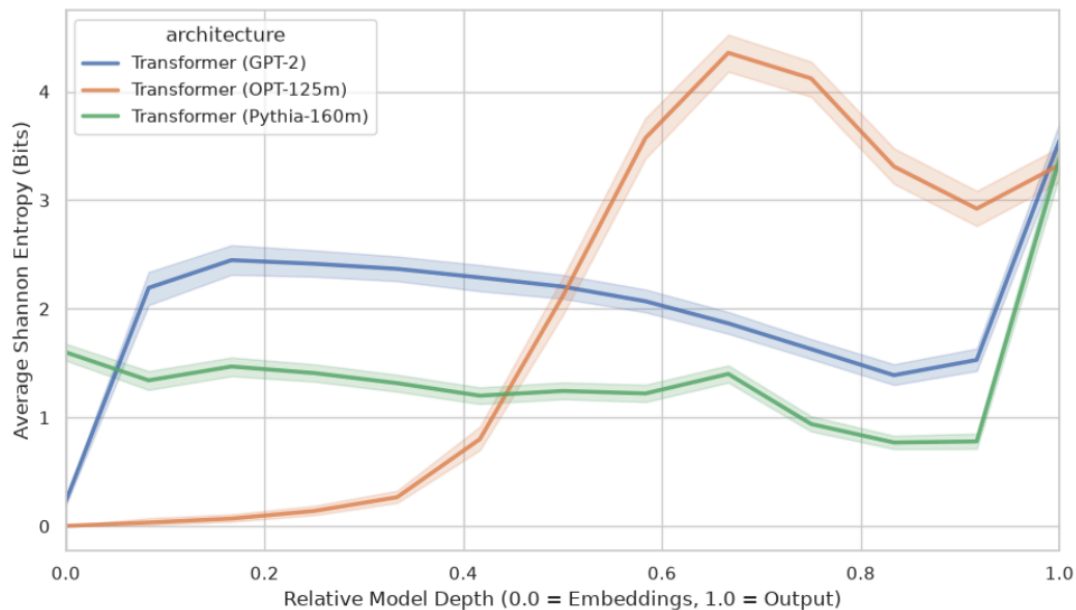


Fig 3: Entropy analysis over 3 distinct transformer-based LLMs

Fig. 3 illustrates the average token entropy across the depth of each layer of OpenAI's GPT-2, Facebook's OPT-125M, and Eletheur AI's Pythia-160M LLM model. We can see that the OPT-125M model's entropy peaks much later compared to the other two models, while the Pythia-160M model starts at a much higher entropy at initial profile. Most importantly, none of the aforementioned models had the same entropy profiles.

Mamba models of the same family type also have similar entropy profiles.

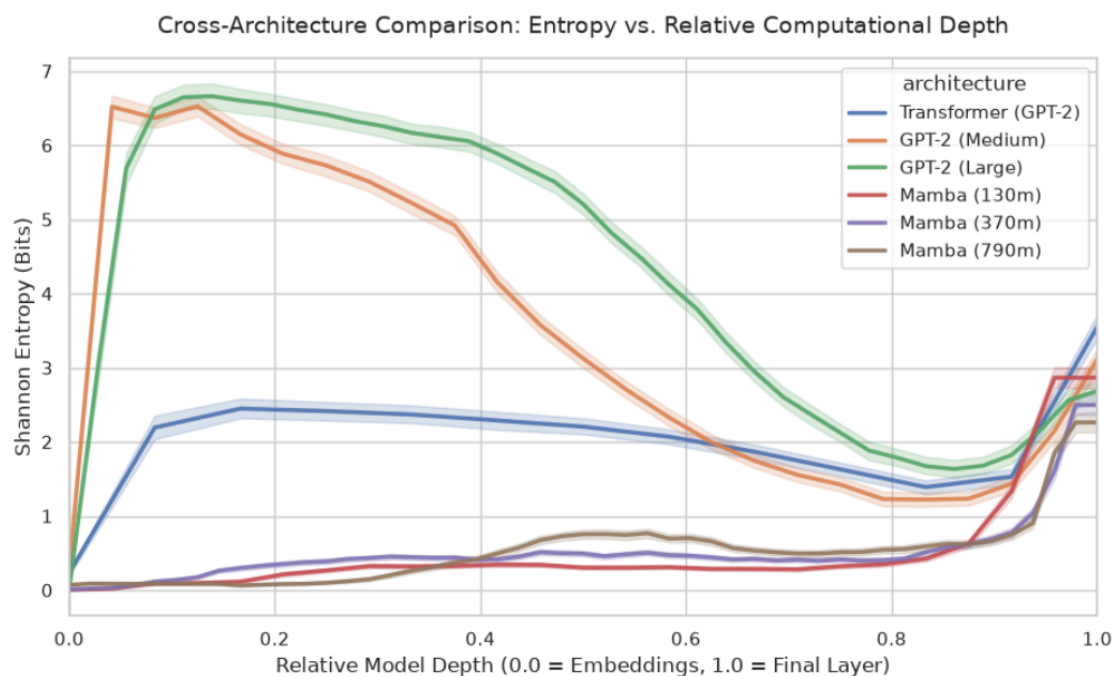


Fig 4: Entropy analysis of GPT-2 (Transformer) and Mamba models (by Tri Dao)

Fig. 4 illustrates that the entropy profile for Mamba models are similar for the same family. It is also important to note that Mamba models do not necessarily have lower entropy compared to Transformer models. Newer and larger models have lower entropy as models become more confident in their prediction. However, this is generally true.

Entropy Analysis for Diffusion LLMs are different from Transformer and Mamba

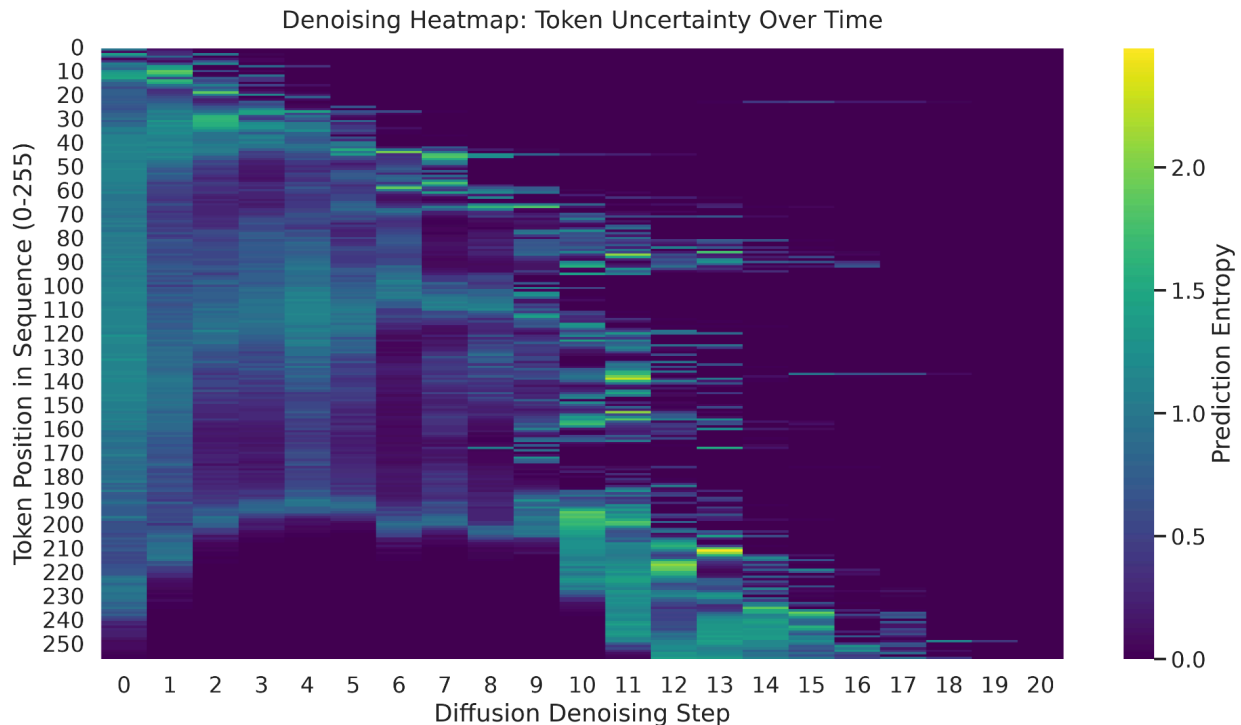


Fig 5: Denoising Heatmap. A visualisation of token uncertainty over time for the Diffusion Gemma model

For the later tokens, token entropy starts at zero, spikes up (bright colour and hence high uncertainty), and then decreases back to zero (low uncertainty).

The phenomena here can be metaphorically described as an “ignorance is bliss” curve, where the model selectively gates information from latter tokens till a certain amount of denoising is done. Information gating can provide a “useful inductive bias for learning representations that achieve better generalization by being robust to noise and spurious correlations.” (Tomar et. al., 2026)

Below states my hypothesis for what is happening:

Phase 1: Pure Static (Low Entropy)

- **When it happens:** Steps 0 to ~5.
- **What is happening:** The canvas is heavily corrupted with 100% random Gaussian noise. The model looks at this absolute static and cannot find a single linguistic pattern.
- **Why entropy is low:** Because the model sees zero signal, it doesn't even try to weigh complex vocabulary words against each other. It just blindly defaults to predicting the safest, most common fallback token (usually an <eos> token, padding, or a period) with extremely high confidence. It isn't confused; it is just defaulting.
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Phase 2: The Signal Emerges (High Entropy)

- **When it happens:** Steps ~10 to ~25 (The bright spots on your heatmap).
- **What is happening:** The diffusion scheduler has removed enough noise that the faint

outline of actual text embeddings starts to appear. The model suddenly realizes, *"Wait, this isn't empty space—this is an English sentence continuing from the prompt!"* * **Why entropy spikes:** Now the model is looking at a "blurry" embedding that could mathematically decode to "apple", "orange", or "banana". Because it is suddenly actively weighing hundreds of valid vocabulary choices against each other, the prediction entropy explodes. This is the moment the model is actually doing the heavy lifting of figuring out the word.

Phase 3: Crystallization (Low Entropy)

- **When it happens:** Steps ~25 to 48.
- **What is happening:** The remaining noise is stripped away.
- **Why entropy drops:** The blurry shape finally crystallizes into the exact continuous embedding for a specific word. The model locks in its final prediction, its confidence shoots to 100%, and the entropy plummets back to zero.

Diffusion models do not denoise linearly

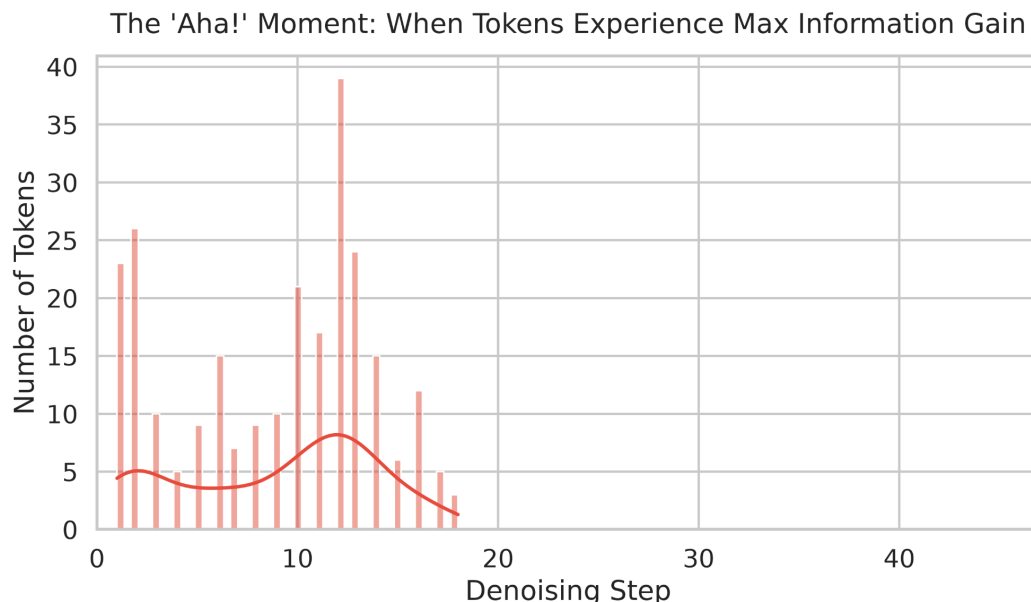


Figure 6: Number of tokens that experience massive entropy drop at each step

The red bars in Fig.6 shows the number of tokens that experience their largest entropy drop at that particular denoising step, while the red line shows the continuous probability distribution of a token experiencing the largest entropy drop in that particular denoising step. Evidently, tokens do not gradually get clearer at a perfectly linear rate. Instead, they coast in a state of high-entropy confusion until a specific threshold of noise is removed, at which point they experience a sudden, sharp drop in uncertainty (the "Aha!" moment).

For Mixture-of-Experts models, router entropy is decoupled from predictive entropy

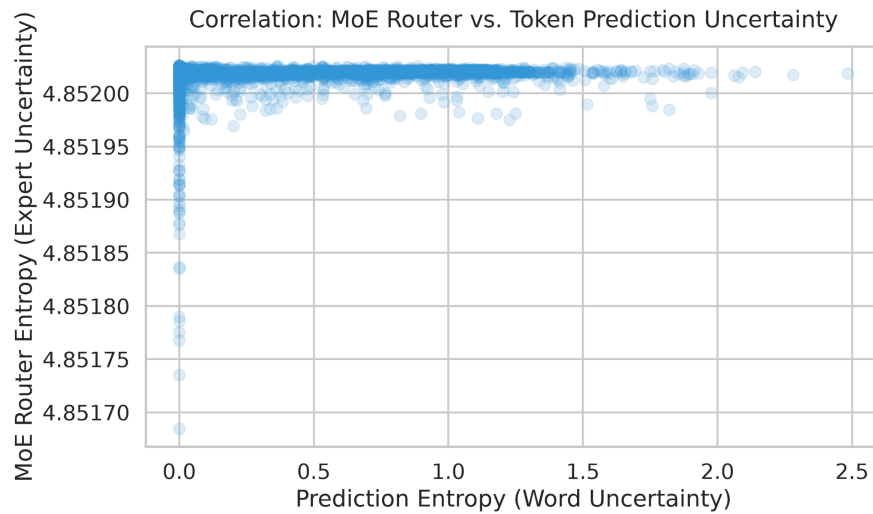


Fig. 7: Router Entropy against Prediction Entropy for Qwen1.5-MoE-A2.7B

Router entropy refers to the level of uncertainty or unpredictability in a system that directs data or computational tasks. From Fig. 7, most data points are scattered along the x-axis at the router entropy of $y = 4.852$. This illustrates that router entropy can be considered to be independent of the predictive entropy.